

MA3236 NONLINEAR PROGRAMMING

Semester 1, AY 2026/2027

Tutorial 3

Review: Lectures 6–7

- **First-order necessary condition.** If $f : \mathcal{D} \rightarrow \mathbf{R}$ is differentiable and $\mathcal{D} \subseteq \mathbf{R}^n$ is open, then

$$\mathbf{x}^* \text{ local optimizer} \implies \nabla f(\mathbf{x}^*) = \mathbf{0}.$$

Such a point is called *stationary*.

- **Second-order necessary condition.** If $f \in C^2(\mathcal{D})$, then

$$\begin{aligned} \mathbf{x}^* \text{ local minimizer} &\implies \nabla f(\mathbf{x}^*) = \mathbf{0}, \quad \mathbf{H}_f(\mathbf{x}^*) \succeq \mathbf{0}, \\ \mathbf{x}^* \text{ local maximizer} &\implies \nabla f(\mathbf{x}^*) = \mathbf{0}, \quad \mathbf{H}_f(\mathbf{x}^*) \preceq \mathbf{0}. \end{aligned}$$

- **Necessary $\neq$ sufficient.**

$$\left. \begin{array}{l} \nabla f(\mathbf{x}^*) = \mathbf{0} \\ \nabla f(\mathbf{x}^*) = \mathbf{0}, \mathbf{H}_f(\mathbf{x}^*) \succeq \mathbf{0} \end{array} \right| \begin{array}{l} \text{candidate only} \\ \text{still not a minimum certificate} \end{array}$$

A stationary point may be a local/global minimum, a local/global maximum, or a saddle point.

- **Hessian test at a stationary point.**

$$\left. \begin{array}{l} \mathbf{H}_f \succ \mathbf{0} \\ \mathbf{H}_f \text{ indefinite} \end{array} \right| \begin{array}{l} \text{strict local min} \\ \text{saddle} \end{array} \quad \left. \begin{array}{l} \mathbf{H}_f \prec \mathbf{0} \\ \mathbf{H}_f \text{ singular semidefinite} \end{array} \right| \begin{array}{l} \text{strict local max} \\ \text{inconclusive} \end{array}$$

- **Unconstrained quadratic.** For $q(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{c}^T \mathbf{x} + \gamma$, $\mathbf{Q} = \mathbf{Q}^T$,

$$\left. \begin{array}{l} \mathbf{Q} \not\succeq \mathbf{0} \text{ or } (\mathbf{Q} \succeq \mathbf{0}, \mathbf{c} \notin \text{Range}(\mathbf{Q})) \\ \mathbf{Q} \succeq \mathbf{0}, \mathbf{c} = \mathbf{Q} \mathbf{w} \end{array} \right| \begin{array}{l} \inf q = -\infty \\ \arg \min q = -\mathbf{w} + \text{Null}(\mathbf{Q}) \end{array}$$

- **Candidate search and classification.**

$$\nabla f(\mathbf{x}) = \mathbf{0} \longrightarrow \text{necessary-condition filters} \longrightarrow \left\{ \begin{array}{l} \text{compare } f\text{-values} \\ \text{sufficient-condition certificates} \\ \text{Singular Hessian treatments} \end{array} \right.$$

For a global conclusion, also prove the well-posedness; the stationary set may be infinite.

- **Domain warning.** $\nabla f(\mathbf{x}^*) = \mathbf{0}$ is an interior condition. A boundary optimizer need not be stationary.

1. Let

$$\mathcal{D} = \{\mathbf{x} \in \mathbf{R}^3 : x_1^2 + x_2^2 + x_3^2 < 4\}$$

and consider minimizing on $\mathcal{D}$ the function

$$f(\mathbf{x}) = \exp(x_1^2 - x_1) + x_2^2 - 4x_2 + 5x_3^2.$$

Does this problem have an optimal solution? Justify your answer using the first-order necessary condition.

Solution. Because $\mathcal{D}$ is open, any local minimizer in $\mathcal{D}$ must be stationary. Here

$$\nabla f(\mathbf{x}) = \begin{pmatrix} (2x_1 - 1) \exp(x_1^2 - x_1) \\ 2x_2 - 4 \\ 10x_3 \end{pmatrix}.$$

The only solution of $\nabla f(\mathbf{x}) = \mathbf{0}$ in $\mathbf{R}^3$ is

$$\hat{\mathbf{x}} = (1/2, 2, 0)^T.$$

But $\|\hat{\mathbf{x}}\|^2 = 17/4 > 4$, so $\hat{\mathbf{x}} \notin \mathcal{D}$. Thus f has no stationary point in $\mathcal{D}$ and hence no local or global minimizer there.

2. Consider

$$q(x_1, x_2) = x_1^2 + 3x_1x_2 + 5x_2^2 + x_1 - 4x_2.$$

Write q as $\frac{1}{2}\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{c}^T \mathbf{x}$. Decide whether a global minimizer exists, whether it is unique, and find the minimizer and minimum value.

Solution. Here

$$\mathbf{Q} = \begin{pmatrix} 2 & 3 \\ 3 & 10 \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 1 \\ -4 \end{pmatrix}.$$

The leading principal minors are 2 and $20 - 9 = 11$, so $\mathbf{Q} \succ 0$. Thus q has a unique global minimizer, determined by

$$\mathbf{Q} \mathbf{x}^* = -\mathbf{c}.$$

Since $\mathbf{Q}(-2, 1)^T = (-1, 4)^T = -\mathbf{c}$,

$$\mathbf{x}^* = (-2, 1)^T, \quad q(\mathbf{x}^*) = -3.$$

3. Consider

$$f(\mathbf{x}) = \ln(1 + x_1^2) + x_1^2 - 2x_1x_2 + 4x_2^2 + x_3^4 - 8x_3^3 + 16x_3^2, \quad \mathbf{x} \in \mathbf{R}^3.$$

(a) Find all stationary points.

(b) Determine all global minimizers of f over $\mathbf{R}^3$.

Solution. The first two stationarity equations are

$$\frac{2x_1}{1+x_1^2} + 2x_1 - 2x_2 = 0, \quad -2x_1 + 8x_2 = 0.$$

The second gives $x_2 = x_1/4$, and substitution into the first gives

$$x_1 \left(\frac{2}{1+x_1^2} + \frac{3}{2} \right) = 0.$$

Hence $x_1 = x_2 = 0$. The third stationarity equation is

$$4x_3(x_3 - 2)(x_3 - 4) = 0.$$

Thus the stationary points are

$$(0, 0, 0), \quad (0, 0, 2), \quad (0, 0, 4).$$

For the global conclusion, rewrite

$$f(\mathbf{x}) = \ln(1+x_1^2) + (x_1-x_2)^2 + 3x_2^2 + x_3^2(x_3-4)^2.$$

Every term is nonnegative, and all four terms vanish exactly at $(0, 0, 0)$ and $(0, 0, 4)$. Hence these are precisely the global minimizers, and the global minimum value is 0.

4. Consider

$$g(x_1, x_2, x_3) = x_1^6 + 5x_2^4 + x_3^2 - 3x_1x_2x_3^2.$$

- (a) Verify that the origin is stationary and that $\mathbf{H}_g(\mathbf{0})$ is positive semidefinite but singular.
- (b) Classify the origin without relying on the Hessian test.
- (c) Is this local conclusion also global?

Solution. Direct differentiation gives

$$\nabla g(\mathbf{0}) = \mathbf{0}, \quad \mathbf{H}_g(\mathbf{0}) = \text{diag}(0, 0, 2) \succeq 0,$$

so the Hessian test is inconclusive. If $\|\mathbf{x}\| < 1/2$, then

$$3|x_1x_2| \leq \frac{3}{2}(x_1^2 + x_2^2) < \frac{3}{8}.$$

Therefore

$$g(\mathbf{x}) = x_1^6 + 5x_2^4 + (1 - 3x_1x_2)x_3^2 \geq x_1^6 + 5x_2^4 + \frac{5}{8}x_3^2 > 0$$

for every nonzero $\mathbf{x}$ in this ball. Thus the origin is a strict local minimizer. It is not global because

$$g(1, 1, t) = 6 - 2t^2 \longrightarrow -\infty.$$